

OmniPitch 2026

The GenAI-Powered Digital Twin & Command Center for the FIFA World Cup 2026

The Problem: Matchday Chaos at Scale

- **Massive Scale:** 80,000 fans enter a stadium simultaneously.
- **Information Silos:** Fans, volunteers, and organizers use disconnected tools.
- **Accessibility & Safety:** Real-time hazards, sensory overloads, and mobility constraints are hard to manage dynamically.
- **Communication Barriers:** Diverse international fans lacking real-time support.

✦ **The Solution: OmniPitch**

A unified, real-time 3D AI Command Center bridging physical infrastructure and digital intelligence.

- **Holographic 3D Digital Twin:** Renders live telemetry in the browser.
- **Google Gemini 2.5 Flash:** Multimodal AI for instant decision-making.
- **Supabase Real-time:** WebSockets for sub-second synchronization.

■ Fan Experience: Hyper-Personalized Safety

- **AI Copilot:** Natural language navigation and support in EN, ES, FR, and DE.
- **Accessibility-First:** Step-free routing and dynamic **Quiet Zone** finding based on real-time crowd density.
- **Proactive Alerts:** Immediate rerouting if a gate becomes critical or a hazard is detected.



Command Center: Organizer & Volunteer Superpowers

- **Global Throughput Heatmaps:** Visualizes gate flows and crowd densities live on the 3D pitch.
- **Vision Triage (Volunteers):** Upload incident photos → Gemini AI assigns severity → Automated dispatch.
- **AI Vibe Engine:** Translates gate delays and crowding into a live Fan Sentiment Score.

Security & Zero Hallucination Architecture

- **Rules-First AI:** Gemini *never* invents routes. Our deterministic Decision Engine resolves facts first; Gemini only handles natural phrasing.
- **Prompt Injection Defense:** User inputs are strictly sandboxed and sanitized.
- **100% Offline Fallback:** Zero-credential boot capability keeps the stadium running safely even if APIs go down.

⚡ Tech Stack & Extreme Performance

- **Frontend Engine:** Vue 3 + Pinia + Vite (Builds perfectly in under 3 seconds!)
- **3D Rendering:** Three.js `InstancedMesh` handles 80,000 seats in a *single draw call* at 60 FPS.
- **AI & Data:** Gemini 2.5 Flash + Supabase WebSockets.
- **Deployment:** Vercel Serverless with strict Rate Limiting and Security Headers.

Business Model & Roadmap

- **MVP (Now):** Single venue AI Command Center live for World Cup 2026.
- **V2 (+6 months):** Hardware IoT integration replacing simulated telemetry.
- **V3 (+12 months):** B2B SaaS Platform — \$50K/year per venue.
- **V4 (+18 months):** Predictive crowd modeling using historical FIFA data.
(50 FIFA-affiliated stadiums = \$2.5M ARR potential)

 **Thank You!**

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Engineering the future of stadium operations.

Live Demo: omnipitch-2026.vercel.app